

CROSS-CUTTING · EXAM STRATEGY - ALL DOMAINS

Confusion Pairs

The service pairs that look alike and decide marks.

COVERS ALL DOMAINS SCENARIOS 8 MODULE 16 of 22

The exam rarely asks 'what is X'. It asks you to choose between **two things that both nearly work.**

Availability and data pairs

PAIR	THE SEPARATING SENTENCE
Multi-AZ vs read replica	Multi-AZ is availability. Read replicas are read scaling.
Versioning vs Object Lock	Versioning recovers deletes. Object Lock prevents them, even by root.
Backup vs durability	Eleven nines does not protect you from deleting the object yourself.
CRR vs AWS Backup	Replication copies objects. AWS Backup applies a policy across services.
Global tables vs Aurora global	DynamoDB multi-active. Aurora one writer with fast promotion.

ON THE EXAM

Multi-AZ versus read replica is the single most repeated confusion on this exam. Over-learn it.

Security and identity pairs

PAIR	THE SEPARATING SENTENCE
CloudTrail vs Config	CloudTrail records actions. Config records state over time.
Macie vs Inspector vs GuardDuty	Macie reads objects. Inspector scans workloads. GuardDuty reads logs.
SCP vs permissions boundary	SCP caps an account. A boundary caps what a principal can create.
Security group vs NACL	Stateful and allow-only, versus stateless and able to deny.
Secrets Manager vs Parameter Store	Rotation costs money. Configuration is free.
Gateway vs interface endpoint	S3 and DynamoDB, free. Everything else, charged.

ON THE EXAM

If a question needs one specific IP address blocked, only a NACL can do it. Security groups have no deny.

Performance and delivery pairs

PAIR	THE SEPARATING SENTENCE
CloudFront vs Global Accelerator	CloudFront caches HTTP. Global Accelerator improves the path, any protocol.
CloudFront vs Transfer Acceleration	CloudFront is downloads. Transfer Acceleration is uploads.
Firehose vs Data Streams	Firehose delivers with no code. Data Streams lets you process and replay.
Athena vs Redshift	Infrequent ad-hoc versus frequent complex analytics.
DataSync vs Storage Gateway	DataSync transfers. Storage Gateway gives ongoing local access.
Reserved vs provisioned concurrency	Reserved limits. Provisioned warms.
Batch vs EMR	Many independent jobs versus one job spread across a cluster.

ON THE EXAM

Reserved and provisioned concurrency is the nastiest pair here, because the names give you no help at all.

An e-commerce database on Amazon RDS is slowing down because a new recommendations feature runs heavy read queries against it during peak shopping hours. The business is satisfied with the database's current availability and has not raised any concern about zone failures.

What should the solutions architect implement?

- ☐ A Convert the database to a Multi-AZ deployment
- ☐ B Create a read replica and point the recommendations feature at it
- ☐ C Enable automated backups with a shorter retention period
- ☐ D Create a cross-Region read replica for the recommendations feature

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A Multi-AZ

Improves availability, which the question says is not a concern, and the standby serves no read traffic at all.

C Backup retention

Backups are unrelated to query performance.

B Read replica in Region

Read scaling is precisely what read replicas do. The heavy queries move off the primary.

D Cross-Region replica

Would work but adds cross-Region transfer cost and latency for a problem entirely contained in one Region.

THE SIGNAL

The question explicitly removes the availability concern. That sentence exists to eliminate Multi-AZ.

An e-commerce database on Amazon RDS serves a steady, moderate query load that the instance handles comfortably.

During a recent Availability Zone impairment the database was unreachable for three hours, costing significant revenue. The business requires that a future zone failure causes only a brief interruption, with no application changes.

What should the solutions architect implement?

- ☐ A Convert the database to a Multi-AZ deployment
- ☐ B Create a read replica in a second Availability Zone
- ☐ C Take more frequent snapshots to reduce recovery time
- ☐ D Increase the instance size so it can absorb more load

◆ PAUSE AND DECIDE

What should the solutions architect implement?

- A

Multi-AZ
Synchronous standby in another zone, automatic failover, and the endpoint name is unchanged so no application change is needed.
- B

Read replica in another AZ
The classic wrong answer. Promotion is manual, replication is asynchronous, and the endpoint changes, so the application must be modified.
- C

More frequent snapshots
Improves the recovery point but restoring from a snapshot still takes far longer than a brief interruption.
- D

Larger instance
The question says the current load is handled comfortably. Size is not the issue.

THE SIGNAL

Identical setup, different failure. One sentence about availability flipped the answer from read replica to Multi-AZ.

A compliance team must be able to answer, for any date in the past eighteen months, whether a particular S3 bucket had public access blocked and encryption enabled. They also want to know who changed those settings and when.

Which combination provides both? (Choose two.)

- ☐ A AWS Config with the configuration timeline for the bucket
- ☐ B AWS CloudTrail event history and log files in S3
- ☐ C Amazon CloudWatch Logs Insights over application logs
- ☐ D Amazon Macie scanning the bucket monthly

◆ PAUSE AND DECIDE

Which combination provides both? (Choose two.)

- A** AWS Config timeline
Config records resource state over time, so it can answer what the bucket's settings were on any given date.
- B** CloudTrail logs
CloudTrail records the API calls, giving the identity that made each change and the time it happened.
- C** CloudWatch Logs Insights
Queries log data you send it. Application logs say nothing about bucket configuration.
- D** Macie
Finds sensitive data inside objects. It does not report on bucket configuration or who changed it.

THE SIGNAL

This question deliberately needs both halves of the pair. State over time is Config; who did it is CloudTrail.

A company delivers a large library of software installers to customers worldwide. The same files are downloaded many thousands of times each day, and the outbound data transfer bill is the largest line on the account.

All traffic is HTTPS.

Which service should be used?

- ☐ A Amazon CloudFront
- ☐ B AWS Global Accelerator
- ☐ C S3 Transfer Acceleration
- ☐ D Route 53 latency-based routing

◆ PAUSE AND DECIDE

Which service should be used?

- A

 CloudFront
Highly cacheable files downloaded repeatedly, over HTTPS, with egress cost as the concern. Every element matches.
- B

 Global Accelerator
Improves the path but caches nothing, so every download still comes from the origin and the transfer bill is unchanged.
- C

 Transfer Acceleration
Accelerates uploads into S3. These are downloads.
- D

 Latency-based routing
Chooses a Region at DNS time. With one origin there is nothing to choose between, and it caches nothing.

THE SIGNAL

Cacheable content served repeatedly means CloudFront. Now watch what happens when the protocol changes.

A company operates a real-time telemetry service. Devices in the field send a continuous UDP stream to endpoints in three Regions. Device firmware has the destination addresses hard-coded and cannot be updated in the field.

If a Region becomes unhealthy, traffic must move to another within seconds.

Which service should be used?

- ☐ A Amazon CloudFront
- ☐ B AWS Global Accelerator
- ☐ C Route 53 failover routing with a sixty second TTL
- ☐ D An Application Load Balancer in each Region

◆ PAUSE AND DECIDE

Which service should be used?

- A

 CloudFront
HTTP and HTTPS only. It cannot carry a UDP stream, and telemetry is not cacheable anyway.
- B

 Global Accelerator
UDP supported, two static anycast IPs that suit hard-coded firmware, and failover in seconds independent of DNS.
- C

 Route 53 failover
Hard-coded IPs mean the devices never perform a DNS lookup, so DNS failover cannot reach them at all.
- D

 ALB per Region
ALBs handle HTTP only, and this provides no mechanism for moving traffic between Regions.

THE SIGNAL

The mirror of scenario four. UDP, hard-coded addresses and seconds-level failover flip the answer to Global Accelerator.

A team runs a Lambda function triggered by API Gateway for a customer-facing checkout page. Customers complain that the first request after a quiet period takes almost three seconds, while subsequent requests are fast.

The downstream systems have ample capacity and are not under strain.

What should the solutions architect configure?

- ☐ A Reserved concurrency on the function
- ☐ B Provisioned concurrency on the function
- ☐ C A larger memory allocation and a shorter timeout
- ☐ D An SQS queue between API Gateway and Lambda

◆ PAUSE AND DECIDE

What should the solutions architect configure?

A Reserved concurrency

Caps how many copies run. It does nothing about start-up latency, and could make things worse by throttling.

C More memory, shorter timeout

More memory can shorten cold starts slightly but does not remove them, and a shorter timeout risks failing slow requests.

B Provisioned concurrency

Keeps execution environments initialised and warm, which is exactly what removes the cold-start delay described.

D SQS between the layers

Makes checkout asynchronous, which a customer waiting for order confirmation cannot accept.

THE SIGNAL

'Slow after a quiet period, fast afterwards' is the textbook description of a cold start.

A regulator requires that transaction records be retained for ten years and be provably impossible to alter or delete during that period, including by administrators and the account root user.

Separately, the operations team wants protection against staff accidentally overwriting current working documents in a different bucket.

Which pairing correctly meets each requirement?

- | | |
|---|---|
| <p>A Object Lock compliance mode for the records; versioning for the working documents</p> | <p>B Versioning for the records; Object Lock governance mode for the working documents</p> |
| <p>C Object Lock governance mode for both</p> | <p>D A bucket policy denying deletes for the records; MFA Delete for the working documents</p> |

◆ PAUSE AND DECIDE

Which pairing correctly meets each requirement?

- | | |
|---|---|
| <p>A Compliance mode, then versioning</p> <p>Compliance mode admits no override, satisfying 'including root'. Versioning makes accidental overwrites recoverable without blocking normal work.</p> | <p>B Versioning, then governance mode</p> <p><i>Exactly backwards. Versioning cannot prevent deletion, and governance mode would obstruct ordinary editing of working documents.</i></p> |
| <p>C Governance mode for both</p> <p><i>Governance mode allows privileged users to override retention, which fails the 'including root user' requirement.</i></p> | <p>D Bucket policy, then MFA Delete</p> <p><i>A bucket policy can be edited by whoever holds the permission, so it provides no immutability guarantee.</i></p> |

THE SIGNAL

Compliance mode allows nobody to override. Governance mode allows privileged override. 'Including root' selects compliance.

A retailer must ingest point-of-sale events from twelve hundred shops. Three separate teams need the same events: a real-time fraud engine that must react within two seconds, an inventory service, and an analytics team that periodically reprocesses the last seven days of events after changing its logic.

No event may be missed by any consumer.

Which service should carry the events?

- ☐ A Amazon Kinesis Data Firehose delivering to S3
- ☐ B Amazon Kinesis Data Streams
- ☐ C A single Amazon SQS standard queue
- ☐ D Amazon SNS with email subscriptions per team

◆ PAUSE AND DECIDE

Which service should carry the events?

A Firehose to S3

Buffers for at least sixty seconds, which breaks the two-second fraud requirement, and it delivers rather than serving multiple live consumers.

C Single SQS queue

A message is consumed once, so whichever team reads it first denies it to the other two.

B Kinesis Data Streams

Multiple consumers read the same records independently, latency is sub-second, and retention allows the analytics team to replay the last seven days.

D SNS to email

SNS does not store, so nothing can be replayed, and email is not an ingestion mechanism for high-volume events.

THE SIGNAL

Replay is the discriminator. Only Data Streams retains records for consumers to re-read.

Module 13 takeaways

- ◆ **Look for the sentence that eliminates.** 'Availability is not a concern' exists to remove Multi-AZ.
- ◆ **Multi-AZ is availability. Read replicas are read scaling.** Over-learn this one.
- ◆ **CloudFront caches. Global Accelerator does not.** Protocol and cacheability decide.
- ◆ **Reserved concurrency limits. Provisioned concurrency warms.**
- ◆ **Replay means Kinesis Data Streams.** 'Including the root user' means Object Lock compliance mode.

ON THE EXAM

Next: how to read a question under time pressure, and what the qualifier words really mean.